

N.S. LABORATORY SDN. BHD.

TIMISIN, 250mg/ml, Solution for use in drinking water

Active Ingredient:

Each ml contains

Tilmicosin (as phosphate)-----250mg

Preservative:

Each ml contains

Methyl paraben-----1mg

Product Description:

Clear yellow to amber coloured solution

Dosage Form:

Solution for use in drinking water

Pharmacodynamics:

Tilmicosin is a semi-synthetic antibiotic of the macrolide group and is believed to affect protein synthesis. It has bacteriostatic action but at high concentrations it may be bactericidal. This antibacterial activity is predominantly against Gram-positive bacteria although some gram-negative bacteria and mycoplasmas of bovine, porcine, ovine and avian origin are also affected by this drug. Scientific evidence suggests that macrolides act synergistically with the host immune system. Macrolides appear to enhance phagocyte killing of bacteria. Tilmicosin has been shown to inhibit *in vitro* the replication of the Porcine Reproductive and Respiratory Syndrome virus in alveolar macrophages in a dose dependent fashion. Cross-resistance between tilmicosin and other macrolides and lincomycin has been observed. Macrolides inhibit protein synthesis by reversibly binding to the 50S ribosomal subunit. Bacterial growth is inhibited by induction of the separation of peptidyl transfer RNA from the ribosome during the elongation phase. Ribosomal methylase, encoded by the erm gene, can precipitate resistance to macrolides by alteration of the ribosomal binding site.

The gene that encodes for an efflux mechanism, mef, also brings about a moderate degree of resistance. Resistance is also brought about by an efflux pump that actively rids the cells of the macrolide. This efflux pump is chromosomally mediated by genes referred to as acrAB genes.

Pharmacokinetics:

Whilst blood concentrations of tilmicosin are low, there is pH-dependent macrophage accumulation of tilmicosin in inflamed tissues.

Indication:

Pigs: For the treatment and metaphylaxis of respiratory disease in pig herds, associated with *Mycoplasma hyopneumoniae*, *Pasteurella multocida*, *Actinobacillus pleuropneumoniae* and other organisms susceptible to tilmicosin.

Chickens: For the treatment and metaphylaxis of respiratory disease in chicken flocks, associated with *Mycoplasma gallisepticum* and *M. synoviae*.

Turkeys: For the treatment and metaphylaxis of respiratory disease in turkey flocks, associated with *Mycoplasma gallisepticum* and *M. synoviae*.

Calves: For the treatment and metaphylaxis of bovine respiratory disease, associated with *Mannheimia haemolytica*, *Pasteurella multocida*, *Mycoplasma bovis*, *M. dispar* and other organisms susceptible to tilmicosin.

The presence of the disease in the group/flock must be established before the product is used.

Mode of Administration:

Oral administration via drinking water.

Recommended Dosage:

For oral use only. Do not inject.

Pigs- To be included in the drinking water to provide a daily dose of 15-20 mg/kg bodyweight for 5 days, which may be achieved by the inclusion of 200 mg tilmicosin per litre (80ml of product per 100 litres of drinking water).

Chickens and Turkeys (except hens producing eggs for human consumption)- To be included in the drinking water at a daily dose of 15-20 mg/kg bodyweight in chickens and 10-27 mg/kg bodyweight in turkeys for 3 days, which may be achieved by the inclusion of 75 mg tilmicosin per litre (30ml of product per 100 litres of drinking water).

Calves- To be included in milk replacer only. 12.5 mg/kg bodyweight and given twice daily for 3-5 consecutive days, which may be achieved by the inclusion of 1ml of product every 20kg body weight.

The uptake of medicated water/milk replacer depends on the clinical condition of the animals. In order to obtain the correct dosage, the concentration of product should be adjusted accordingly. To ensure a correct dosage, bodyweight should be determined as accurately as possible to avoid under-dosing. The required dose should be measured using suitably calibrated measuring equipment. Only sufficient medicated drinking water should be prepared to cover the daily requirements.

Contraindications:

Do not allow horses and other equines access to drinking water containing tilmicosin. Do not use in case of hypersensitivity to tilmicosin or to any of the excipients.

Special Warnings for each Target Species:

Important: Must be diluted before administration to animals.

Pigs, chickens and turkeys: Water consumption should be monitored in order to guarantee adequate dosing. In case water consumption does not match quantities for which recommended concentrations were calculated, the concentration of the product has to be adapted in a way that the recommended dosage will be taken up by the animals or different medication should be considered.

Special Precautions for Use in Animals:

For oral use only. Contains EDTA; do not inject. Severely ill individuals tend to drink

less and may need simultaneous treatment, preferably by parenteral medication. Inappropriate use of the product may increase the prevalence of bacteria resistant to tilmicosin and may decrease the effectiveness of treatment with tilmicosin related substance. Use of the product should be based on susceptibility testing of the bacteria isolated from the animal. If this is not possible, therapy should be based on local (regional, farm level) epidemiological information about susceptibility of the target bacteria. Medicated drinking water or milk replacer should be prepared fresh every 24 hours. Official, national and regional antimicrobial policies should be taken into account when the product is used.

Special Precautions to be taken by the Person Administering the Medicinal Products to Animals:

Tilmicosin may induce irritation. Macrolides, such as tilmicosin, may also cause hypersensitivity (allergy) following injection, inhalation, ingestion or contact with skin or eye. Hypersensitivity to tilmicosin may lead to cross reactions to other macrolides and vice versa. Allergic reactions to these substances may occasionally be serious and therefore direct contact should be avoided. To avoid exposure during preparation of the medicated drinking water, wear overalls, safety glasses, and impervious gloves. Do not eat, drink or smoke when handling this product. Wash hands after use. In the case of accidental ingestion, wash out mouth immediately with water and seek medical advice. In the event of accidental skin contact, wash thoroughly with soap and water. In case of accidental eye contact, flush the eyes with plenty of clean, running water. Do not handle the product if you are allergic to ingredients in the product.

If you develop symptoms following exposure, such as skin rash, you should seek medical advice and show the physician this warning. Swelling of the face, lips and eyes or difficulty in breathing are more serious symptoms and require urgent medical attention.

Interactions with other Medicaments:

None known

Usage during Pregnancy and Lactation:

The safety of tilmicosin has not been established during pregnancy, lactation or lay. Use only in accordance with risk/ benefit assessment by the responsible veterinarian.

Adverse Effects:

In very rare cases, a decrease in water intake has been observed.

Overdose and Treatment:

When pigs are offered drinking water containing 300 or 400 mg tilmicosin /litre (equivalent to 22.5-40 mg tilmicosin /kg bodyweight or 1.5-2 times the recommended concentration) commonly animals exhibit a reduced water intake. Although this has a self-limiting effect on tilmicosin intake, it could, in extreme circumstances, result in dehydration. This can be corrected by the removal of the medicated drinking water and replacement with fresh unmedicated water.

No symptoms of overdose have been seen in chickens given drinking water containing levels of tilmicosin up to 375 mg/litre (equivalent to 75-100 mg tilmicosin /kg bodyweight or 5 times the recommended dose) for 5 days; daily treatment with 75 mg tilmicosin /litre (equivalent to the maximum recommended dose) for 10 days resulted in a reduction in faecal consistency.

No symptoms of overdose have been seen in turkeys given drinking water containing levels of tilmicosin up to 375 mg/litre (equivalent to 50-135 mg tilmicosin /kg bodyweight or 5 times the recommended dose) for 3 days; daily treatment with 75 mg tilmicosin /litre (equivalent to the maximum recommended dose) for 6 days also produced no symptoms of overdose.

No symptoms of overdose, with exception of a slight decrease in the milk consumption, have been seen in calves given twice daily doses 5 times the maximum recommended dose or for twice the maximum recommended duration of treatment.

Withdrawal Periods (Meat and Offal):

Pigs- 14 days

Chickens- 12 days

Turkeys- 19 days

Calves- 42 days

Not authorised for use in animals producing milk for human consumption.

Not authorised for use in birds producing eggs for human consumption.

Do not use within 2 weeks of the start of the laying period.

Storage Condition:

Store in cool, dark place(< 30°C). Protect from direct sunlight.

Shelf Life:

36 months

Shelf Life after Opening:

Discard any unused solution immediately after opening

Shelf Life after Dilution/Reconstitution:

24 hours

Packaging Available:

1L/bottle; 12 bottles/carton; 12kg/carton

Manufacturer and Registration Holder:

N.S. Laboratory Sdn. Bhd.

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